Simulation Instruction Document

Gameplay description

In Six Day Garden, the player plants and cares for a Universe Flower. If cared for properly, the Universe Flower can fully bloom by day 6. Each day, the flower must be watered only once, free of aphids, free of excess pest spray, and have compost to eat. The player determines when and how many times the plant is watered, sprayed for pests, given compost, and when to move on to the next day.

-  Water action:
  - This is a blue button. When this is clicked, a blue dot is added to the plant to indicate it has been watered that day. If the plant is watered more than once each day, the plant health will decrease with each watering.
-  Add compost action:
  - This is a yellow button. When this is clicked, a piece of compost will be added to the soil. The compost pieces are: a green apple, a banana, an orange, a pear, a blueberry, and a strawberry. One piece will “be consumed” (disappear) each day. If the player tries to add a seventh piece of compost, they will be told they don’t have anymore. If a day passes without compost in the soil for the plant to eat, the plant health will go down.

    

-  Pest spray action:
  - This is a red button. When this is clicked, one aphid will be removed from the plant. If there are no aphids, the plant health will decrease with each spray.
-  Sleep action:
  - This is a green button. This causes the day counter to go up, lack of compost decreases plant health by 1, a piece of compost is eaten, the water count is updated to 0, and any remaining aphids decrease plant health.

Win/lose scenarios

There are 3 possible winning scenarios:

- Best scenario: To win, the player must reach day 6 with the plant in stage 3. When the player does this, they win first prize in the Horticulture Fair Competition.
- Second best scenario: When the player reaches day 6 with a stage 2 plant, they win 3rd place in the competition and are told to try for first place next year.
- Third best scenario: When the player reaches day 6 with a stage 1 plant, they receive a participation ribbon and are told to try for a podium spot next year.

There are 5 possible losing scenarios:

- Not enough water: If lack of water brings plant health to 0, the player is told that their plant did not receive enough water.
- Too much water: If excess water brings plant health to 0, the player is told that they've overwatered their plant.
- Too much pest spray: If excess pest spray brings plant health to 0, the player is told that they've sprayed too much pest spray.
- Eaten by aphids: If aphids bring plant health to 0, the player is told that aphids have eaten their plant.
- Not enough compost: If lack of compost brings plant health to 0, the player is told that their plant did not get enough compost.

Game and player avatars

This game is played in first person. The flower is in the center of the screen, the action buttons are on the bottom, and the counters are on the top left. The plant appears in three different stages:

- Stage 1: The plant is a 4 leaf clover.



- Stage 2: The plant has yellow bulbs and one flower blooming.



- Stage 3: The plant has one large yellow flower with a purple center.



On some days, 3 to 5 aphids will be present on the plant. These are small red bugs.

Controls and what they do

All four buttons can be clicked with the mouse or with their corresponding key on the keyboard.

- a = water
- s = add compost
- d = pest spray
- w = sleep